



Local Difference-Based Federated Learning Against Preference Profiling Attacks

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Abstract. The recommendation system based on federated learning has become one of the most popular distributed machine learning technologies, which to some extent protects the privacy and security of users. However, personal privacy information can still be disclosed through privacy inference attacks. In order to resist preference profiling attack (PPA), we propose a differential privacy federated learning recommendation method based on federated neural collaborative filtering. We construct a neural collaborative filtering recommendation method based on federated learning with local differential privacy, which can achieve the effect similar to that of federal neural collaborative filtering recommendation. First, we compared three instances of neural collaborative filtering, i.e., Multi-Layer Perceptron (MLP), Generalized Matrix Factorization (GMF), and Neural Matrix Factorization (NeuMF) combined with federated learning, under the same conditions, in order to select the optimal instance. And then the training parameters varied using Gaussian noise of the gradient of the uninteracted items and Laplace noise of the interacted items. Under the same evaluation indicators, the training loss rate has decreased by 1.61%, and the recommendation hit rate hit_ratio@10 was improved by 3.05%, normalize discounted cumulative gain ndcg@10 was reduced by 2.55%. The experimental results validated the feasibility of this method, indicating that the improvement of the algorithm is effective.

Keywords: Differential Privacy · Federated Learning · Neural Collaborative Filtering · Preference Profiling Attack

1 Introduction

In recent years, due to the popularization of mobile devices such as mobile phones, China's mobile Internet industry has developed rapidly. According to the "Statistical Report on Internet Development in China" released by the China Internet Network Information Center for the 50th time, as of June 2022, the

Internet penetration rate in China has reached 74.4%. The report shows that the number of Internet users in China has reached 1.051 billion, and this scale is still growing steadily. It is particularly worth noting that the number of short video users has shown the most significant growth, reaching 962 million, an increase of 28.05 million compared to December 2021, accounting for 91.5% of the entire Internet user population [1]. In addition to short videos, there are many industries similar to it, such as news. With the rapid development of various industries, massive amounts of information are generated on the Internet every day, and users wanting to find their favorite information among them is tantamount to finding a needle in a haystack. Therefore, a recommendation system emerged as the times require. Through the recommendation system, massive data can be integrated and accurately pushed to users, reducing the cost of data screening for users.

With the promulgation and implementation of the General Data Protection Regulation [2] and other privacy data protection laws and regulations, as well as the enhancement of people's awareness of privacy protection, the privacy security of personal data has been paid more and more attention [3]. However, most of the current recommendation systems run on a central server, and the users' personal information and the interaction information between users and items are also stored on the central server, so the recommendation system faces the problem of user privacy leakage.

In order to solve the problem of privacy leakage faced by the recommendation system, Google proposed the federated learning paradigm [4] so that the user's original data is kept locally, and various privacy protection technologies can be comprehensively used locally on the user side to achieve better the purpose of protecting user privacy [5]. However, the application of the federated learning framework has many limitations, and it is not absolutely safe in practical applications. There are many attack methods that can threaten the data security in the federated learning process [6]. For instance, Li et al. [7] trained a poisoned model with a poisoned data set and replaced the neural pathways of the target model with its neural pathways to achieve poisoning attacks; Wang et al. [8] used toxic adversarial samples for model training and improved learning rate to accelerate the generation of poisoning models, thus improving the attack performance of poisoning attacks. The above two examples are poisoning attacks. In addition, reasoning attacks are the most typical attack methods faced by federated learning, threatening the privacy of users [9]. The difference can be divided into attribute reasoning attack and member reasoning attack [10]. Recently, some scholars have studied various inference attacks that federated learning may suffer, and proposed a new inference attack method, that is, preference profiling attack (PPA) [11]. This attack can analyze the local model gradient uploaded from each user's end thus obtaining private information such as user's preference for items [12], which is a serious threat to the user's privacy and security [13]. Therefore it is important to find a way to defend against the preference PPA of federated learning in practical applications.

To address the above issues, based on the above research ideas, the specific work completed in this paper is as follows:

Contribution1. A neural matrix factorization recommendation model based on federated learning is designed. The training efficiency and recommendation effect of three instances of federated neural collaborative filtering are compared and analyzed through experiments.

Contribution2. An improved differential privacy federated learning recommendation method that resists PPA is designed, and Laplace noise and Gaussian noise are added to the gradient of the local model for processing. The training efficiency and recommendation effect before and after adding differential privacy are compared and analyzed through experiments.

2 Related Works

2.1 Federated Recommendations Face Various Gradient Attacks

Federated learning is not absolutely safe. In order to build a global model for federated learning, users still need to upload local model parameters or gradients. These parameters or gradients are essentially a mapping of the user's local data and contain almost all information about the user's data. Multiple attack models show that by analyzing the parameters or gradients of local models, partial or complete information of the original data used by users for training can be obtained [14].

Truc et al. [15] proposed an Active Membership Inference, in which the server forges malicious parameters and embeds them into the global model to infer whether the target data samples are included in clients' private in the training data. Hu et al. [16] proposed Source Inference Attack (SIA), after determining which data instances are training members in membership inference attack, the attacker can further conduct SIA to identify which client it came from. Chen et al. [16] combined membership inference attack with double threshold function to improve the efficiency of distinguishing members from non-members. Peng et al. [17] combined principal component analysis with membership inference attack in the black-box membership attack scenario to suppress the low migration behavior caused by over-reliance on the model and improve attack efficiency.

In addition, Zhou et al. [18] proposed PPA. Through this attack, the attacker can obtain the user's privacy preferences, such as the favorite or least favorite when shopping online. Products and selfie expressions of photos you like to use, etc.

2.2 Federated Recommendations Face Various Gradient Attacks

The above-mentioned attack methods obtain users' private information by analyzing the gradient. The existing protection gradient research can be divided into gradient protection methods based on homomorphic encryption, gradient protection methods based on secret sharing, and gradient protection methods based on the gradient protection method of differential privacy and the gradient protection method of mixing multiple strategies.

Chen et al. [19] combined federated learning and homomorphic encryption algorithm for the training of electric power data prediction model, so as to protect the gradient of the local model; similarly, Yuan [20] used homomorphic encryption technology to encrypt the vertical federation Local model gradients during learning to avoid privacy leaks. Chen et al. [33] combined additive secret sharing technology to enhance the secrecy of gradient parameters on the basis of using compressed sensing to reduce the communication overhead of the federated learning framework; Wang et al. [21] used secret sharing to a reusable security parameter mask is generated in a way, and a mask is added to the model in the uplink communication of the training process to protect the security of the model gradient.

Liu et al. [22] proposed a secure federated transfer learning framework for neural network models, using homomorphic encryption and secret sharing to preserve privacy. Gao et al. [23] proposed a heterogeneous federated transfer learning method for logistic regression and support vector machine models, also using homomorphic encryption and secret sharing to protect privacy. Since there is no guarantee that the third-party server will not collude with the central server, there is still a risk of privacy leakage.

Xu et al. [24] proposed a federated learning method based on differential privacy-protected knowledge transfer, using differential privacy technology to add noise to the gradient that needs to be transmitted during the knowledge transfer process, realizing privacy-preserving knowledge transfer, thereby reducing privacy indirect leakage possibility. Wang et al. [25] proposed an adaptive tailoring differential privacy federated learning framework, which adds dynamic Gaussian noise to the uploaded model parameters, and then uploads the noise-added model parameters to the server for global model parameter update, ensuring data privacy security during parameter transmission is ensured.

Lin et al. [26] proposed a privacy-preserving federated recommendation algorithm based on data perturbation in order to protect user privacy in federated learning. Collection of ratings. During training, the algorithm improves privacy protection during transmission by mixing real and generated sets and uploading them. Although this method protects the interaction history of user items, it does not do any processing on the gradient in the real interaction set. Attackers can still obtain the user's real ratings for some items through the gradient, and there is a risk of privacy leakage.

It can be seen that the above strategies of federated learning to protect user privacy have certain limitations, and they cannot well defend against preference profiling attack. In order to better protect user privacy and prevent them from being attacked by preference analysis, this paper will combine differential privacy and data perturbation Technology Improvement Based on Federated Learning Neural Collaborative Filtering Recommendation Algorithm.

2.3 Attack Model

The PPA is a white-box attack, which assumes that the central server is malicious and selects samples in each item category and retrains the local model for

a specific user to obtain the local model gradient sensitivity for each item category. The attack uses a selective aggregation mechanism to improve the gradient sensitivity by aggregating a subset of user-side of interest to the central server. It utilizes the attack model as a meta-classifier to predict the target user data preference by taking the extracted sensitivity information of the user uploaded and aggregated models as input.

The process of PPA is shown in Fig. 1. It mainly includes four steps. The first step is that the user trains the model locally and uploads the local model gradient to the central server. The second step is that the central server retrain the model using auxiliary datasets to obtain model sensitivities for each item class. The third step is to use the meta-classifier to predict the user's preferred items. The fourth step is selective aggregation. For users who are interested in the server, select other pairs of preference classes. The model parameters of the most sensitive users are aggregated and sent to the target users.

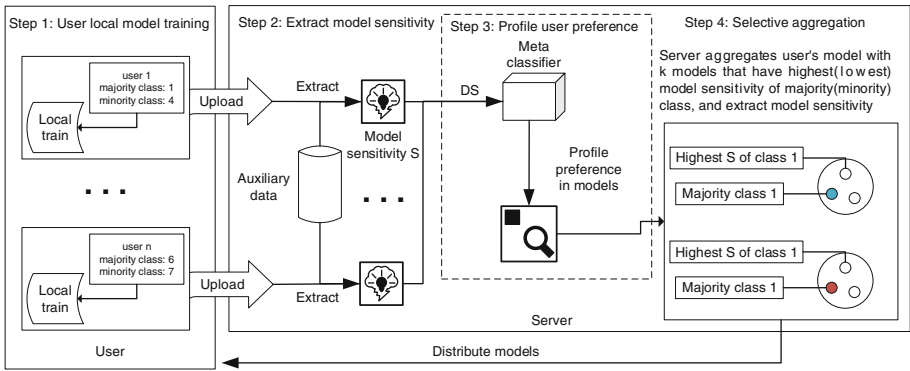


Fig. 1. Preference Profiling Attack Flow

In general, the PPA is an attack against users uploading gradients, and the capture of information such as gradients may lead to privacy leaks has been proved by Li et al. [27] privacy breach.

3 Recommendation Method Based on Improved Differential Privacy Federated Learning

3.1 Neural Matrix Factorization Recommendation Algorithm Based on Federated Learning

Neural Collaborative Filtering (NCF) is a general framework for collaborative filtering based on neural networks proposed by He et al. [28], which consists of five layers. Its input layer is composed of two feature vectors describing users and items respectively, and the embedding layer above the input layer is a fully

connected layer, which projects the sparse representation to the dense vector. By feeding the embedding layer into the hidden layer, which maps the latent vector to the predicted score of the item. Each hidden layer can be different from the previous ones to discover specific latent structures in user-item interaction information. The performance of the model depends on the dimension of the last hidden layer.

He et al. [29] proposed three instances based on NCF, the first is Multi-Layer Perceptron (MLP), the second is Generalized Matrix Factorization (GMF) and the third is Neural Matrix Factorization (NeuMF) which fuses Multi-Layer Perceptron and Generalized Matrix Factorization. The main difference between these three instances is that they have different implementations of the neural collaborative filtering layer. NeuMF embeds user and item features independently for MLP and GMF at the embedding layer, thus avoiding the problem of consistent input matrices leading to poor results. In the collaborative filtering layer, MLP and GMF are mapped separately, and the results of the two models are merged in the last layer of the hidden layer and the predicted scores are output.

The framework of the neural matrix decomposition recommendation algorithm based on federated learning is shown in Fig. 2. The architecture adopted by the federated neural matrix decomposition is the client-central server architecture. A client can be one user or several users, and each client can train a local recommendation model independently. The central server is used to update the parameters of the global model and coordinate and control the entire training process. The aggregation of local model parameters is its core function. The local model parameters of each client are aggregated into new global model parameters.

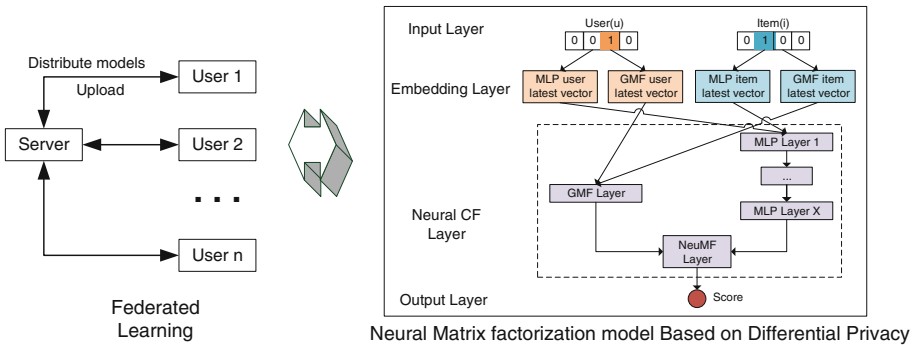


Fig. 2. Neural Matrix Factorization Framework Based on Federated Learning

3.2 Gradient Transfer Model Based on Differential Privacy

Common privacy protection technologies include differential privacy, homomorphic encryption, and secure multi-party computation [32]. Since Zhou [8] and

Algorithm 1. Improved Federated Learning Algorithm based on Differential Privacy**Input:** Client c **Output:** Local model parameters ω_{t+1}^c **Server execution:**

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1: Initialize global model parameter;
2: for every epoch  $t = 1, 2, 3, \dots$  do
3:    $m = \max(p * c, 1)$ ;
4:    $S =$  Randomly selected  $m$  users;
5:   for every client  $c$  belongs to  $S_t$  do
6:      $\omega_{t+1}^c =$  Clients update function  $ClientUpdate(c, \omega_t)$ ;
7:    $\omega_{t+1} = \sum_{c \in S_t} \frac{n_c}{n} \omega_{t+1}^c$ ;

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Clients update function $ClientUpdate(c, \omega_t)$:

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1:  $B =$  Divide  $D_c$  into batches of size  $B$ ;
2: for every client's epoch  $i = 1, 2, 3, \dots, E$  do
3:   for batch do
4:      $\omega_{t+1}^c = \omega_c - \eta \nabla \vartheta(\omega_t; b)$ ;
5: Adding Laplace noise to non-zero parameters:
6:    $\omega_{t+1}^c = Laplace(\omega_{t+1}^c, \varepsilon)$ ;
7: The model parameter for this iteration is  $params = \omega_{t+1}^c$ ;
8: if  $last\_params == \text{None}$  then
9:   The previous iteration's parameter is equal to the current iteration's:
10:   $last\_params = params$ ;
11: else
12:  Calculate the standard deviation sigma between  $last\_params$  and  $params$ ;
13:  Generate  $fake\_params$  of pseudo interactive items using sigma as the standard
    deviation of Gaussian noise;
14:  Add Gaussian noise  $\omega_{t+1}^c = \omega_{t+1}^c + fake\_params$ ;
15:   $last\_params = params$ ;
16: return Local model parameters  $\omega_{t+1}^c$  to server;

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others proved that after the introduction of differential privacy technology, federated learning can reduce the accuracy of PPA to a certain extent, thereby achieving the purpose of protecting user privacy. Differential privacy can be applied to various machine learning algorithms and optimization techniques. In federated learning, differential privacy can be combined with optimization methods such as gradient descent and stochastic gradient descent to provide privacy protection for gradient updates on each device. Therefore, this paper chooses to use differential privacy technology to improve the federated learning algorithm so as to resist PPA and protect user privacy.

Differential privacy generates a series of perturbed data with privacy-preserving properties by randomly perturbing the original data, so as to achieve the purpose of privacy protection [29]. In federated learning, noise can be added on the user's local device, and then only the noised gradients are shared. In this way, even if the malicious server in the PPA obtains this gradient information,

it cannot accurately restore the real gradient of the entire data set, thereby protecting the privacy of individual data. The most common types of differential privacy are Laplace noise and Gaussian noise. These two types of noise have their own advantages. Therefore, this paper comprehensively uses the two types of noise to protect user privacy in gradients.

Because PPA can collect private information such as through interaction history records by the gradient analysis, we employ Laplace noise for numerical queries to prevent PPA from obtaining user-side preferred item information; Gaussian noise with better smoothness and continuity is used to generate random pseudo-interaction item information to prevent the leakage of user item interaction history information. The specific algorithm is shown in Algorithm 1.

Where C is the total number of participant ends, P is the proportion of randomly selected participant ends, S_t the set of randomly selected participant ends, n_c is the number of local training instances, n is the total number of training instances, ω_{t+1}^c the local weight update generated by participant ends, B is the local training minimum batch size, E is the number of rounds of local training, D_c is the data of the participant end, and η is the learning rate of training.

The algorithm is modified on the basis of FedAvg. First, after the training of each client end, a Laplace noise is added to the model parameters to resist PPA and obtain user preference information. The privacy budget of the Laplace noise ϵ is 0.1; then record the model parameters of each round of local training, if it is the first round of training, record the model parameters of this round as the model parameters of the previous round, and upload it to the central server, otherwise calculate the parameters of this round and the previous round. The standard deviation of the parameters is used as the standard deviation of the Gaussian distribution to randomly generate the parameters of the pseudo-interaction items to resist the PPA and obtain the user item interaction history.

3.3 Privacy Analysis

Definition 1. For a function $f : D \rightarrow R^d$, its maximum Manhattan distance output conditional on two neighboring datasets D_1 and D_2 denotes the sensitivity, as shown in Eq. 1 [30]:

$$\Delta f = \max_{(D_1, D_2 \in Q)} \|f(D_1) - f(D_2)\| \quad (1)$$

In the above equation, R denotes the real number space of the mapping and d denotes the query dimension of the function f . The sensitivity Δf is independent of the dataset and related to the function f .

Definition 2. Given neighboring datasets D_1 and D_2 , assume that there is a privacy algorithm F and $\text{Range}(F)$ is all possible outputs of F . Any output of algorithm F on datasets D_1 and D_2 is O ($O \in \text{Range}(F)$), which satisfies the following inequality $\Pr(F(D_1) = O) \leq e^\epsilon \Pr(F(D_2) = O)$, then Algorithm F is said to satisfy the ϵ -differential privacy mechanism, where ϵ is the differential

privacy budget, and the smaller the value, the higher the probability that $F(D_1) = 0$ is equal to $F(D_2) = 0$, and the higher the level of privacy preservation of Algorithm F is.

Theorem 1. For any function $f : D \rightarrow R^d$, if the output result of algorithm F satisfies $F(D) = f(D) + \left\langle \text{Lap}_1(\frac{\Delta f}{\epsilon}), \dots, \text{Lap}_d(\frac{\Delta f}{\epsilon}) \right\rangle$, then it is said that the algorithm F satisfies ϵ -differential privacy [31].

Theorem 2. Postprocessing Resistance Theorem. If $G : N^{|X|} \rightarrow R$ satisfies ϵ -differential privacy and $H : R \rightarrow R'$ is an arbitrary function, then $H \circ G : N^{|X|} \rightarrow R'$ still satisfies ϵ -differential privacy.

Proof. We know from the Laplace noise mechanism that the gradient after adding Laplace noise in Algorithm 1 is $\omega_{t+1}^c \leftarrow \omega_{t+1}^c + \text{Lap}(\frac{\Delta \omega_{t+1}^c}{\epsilon})$, which satisfies Theorem 1, and therefore it satisfies ϵ -differential privacy. From Theorem 2, post-processing the result for ϵ -differential privacy does not weaken its differential privacy protection. Therefore, the gradient after adding Gaussian noise in Algorithm 1 is $\omega_{t+1}^{\tilde{c}} = \omega_{t+1}^c + \text{fake_params}$, and it still satisfies ϵ -differential privacy.

In summary, Algorithm 1 satisfies the ϵ -differential privacy-preserving mechanism and realizes the privacy-preserving function for the gradient in federated learning.

4 Simulation

4.1 Lab Environment

This experiment is carried out under Windows 10 operating system, Intel(R) Core(TM) i5-8250U CPU@ 1.60 GHz 1.80 GHz, no GPU. The software environment is Python3.6, PyTorch1.10.2.

4.2 Evaluation Index

In this experiment, the loss rate loss is used as the evaluation index of training efficiency, and the two classic recommendation system evaluation indexes of Hit Ratio (HR) and Normalized Discounted Cumulative Gain (NDCG) are used to measure the recommendation effect. The formula for calculating the hit rate is as follows:

$$\text{hit_ratio}@k = \frac{\text{Number_of_hits}@k}{GT} \quad (2)$$

In the formula, the denominator is the sum of the test set, and the numerator is the number of the first k recommended items in the recommendation list of each client that belong to the test set. The higher the value, the better the effect of the recommendation system. The calculation formula of $NDCG@k$ is as follows:

$$DCG@k = \sum_i^k \frac{r(i)}{\log_2(i+1)} \quad (3)$$

$$IDCG@k = \sum_i^{|REL|} \frac{r(i)}{\log_2(i+1)} \quad (4)$$

$$NDCG@k = \frac{DCG@k}{IDCG@k} \quad (5)$$

In the formula, $r(i)$ is the correlation score of each item in the data set, DCG is the recommendation effect for a certain user, IDCG is the real recommendation effect score for a certain user, and $|REL|$ represents the items in the recommendation list from large to small according to the recommendation score. Arranged in sequence, a collection of the first k items. The normalized score NDCG is obtained by dividing IDCG by each user's DCG. The higher the value of NDCG, the better the effect of the recommendation system.

4.3 Experimental Dataset

In this experiment, we used the MovieLens-1M dataset as the experimental data. The dataset contains personal information of 6040 users, information of 3952 movies and ratings of the movies by these users. Among them, the number of movies rated by each user is more than 20. In order to facilitate subsequent operations, we serialize and map the IDs of users and movies. This experiment only uses the data in the rating file ratings.dat. More specifically, we only use the rating data of the top 250 users on the movie. The data format is "UserID::MovieID::Rating::Timestamp", and the specific meanings are shown in the Table 1:

Table 1. MovieLens-1M Data Fields

Field Name	Meaning
UserID	ranging from 1 to 6040
MovieID	ranging from 1 to 3952
Rating	Movie rating, five-point scale, ranging from 1 to 5, all integers
Timestamp	Point in time for user ratings

4.4 Superiority Analysis

The purpose of this experiment is not to present more effective recommendations, but to compare three instances of NCF after federated learning, and thus to show the advantages of choosing neural matrix decomposition over the other two instances. The specific comparison models are:

FedGMF (Federated Generalized Matrix Factorization): Uses user-item data for matrix decomposition, transform the high-dimensional rating matrix into the product of the status feature matrix, then use the SGD method to converge the model to obtain the user's local model, and finally obtain the global model through the FedAvg method Model.

FedMLP (Federated Multi-layer Perceptron): Uses user-item data to train the interaction between the user and the latent features of the item, and then predicts it through the activation function ReLU.

FedNeuMF (Federated Neural Matrix Factorization): Through generalized matrix factorization and multi-layer perceptron fusion to jointly train user-item linear and nonlinear interactions for prediction. The experimental results are shown in Table 2 and Fig. 3 below:

Table 2. Indicators of the Three Federated NCF Models Under the Movielsens-1M Dataset

Recommendation Model	loss	hit_ratio@10	ndcg@10
FedGMF	0.0962	0.7565	0.6960
FedMLP	0.0289	0.8881	0.9272
FedNeuMF	0.0186	0.8955	0.9325

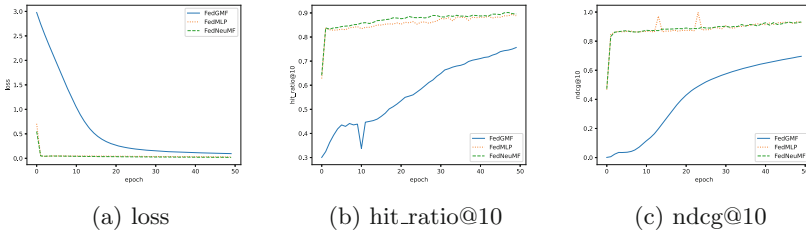


Fig. 3. Changes of Three Instances under Movielsens-1M

4.5 Effectiveness Analysis

This experiment was carried out on the MovieLens-1M dataset, and the recommendation effect and training efficiency of the FedNeuMF algorithm before and after adding differential privacy were compared. The experimental results are shown in Table 3:

Table 3. Indicators Before and After Adding Differential Privacy under The Movielsens-1M

Recommendation Model	loss	hit_ratio@10	ndcg@10
FedNeuMF	0.0186	0.8955	0.9325
DP-FedNeuMF	0.0183	0.9228	0.9087

The privacy budget epsilon for this experiment was 0.1 and the rest of the parameters were the same as in the previous experiment. Compared with the

final value of training, it can be seen from Table 3 that on the Movielens-1M dataset, the loss of DP-FedNeuMF is lower than that of FedNeuMF. The hit rate hit_ratio@10 of DP-FedNeuMF is 3.05% higher than that of FedNeuMF; the cumulative gain of normalized loss ndcg@10 of DP-FedNeuMF is 2.55% lower than that of FedNeuMF.

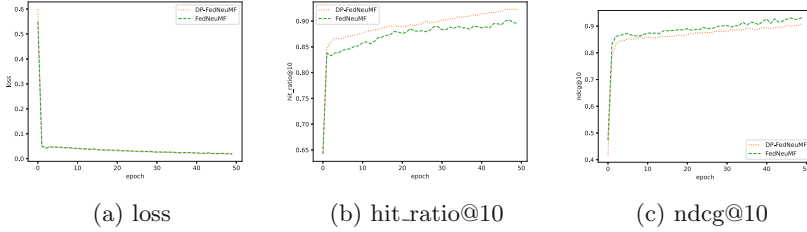


Fig. 4. Changes in Indicators Before and After Adding Differential Privacy

From Fig. 4, we can see that with the increase of training times, the hit_ratio@10 and ndcg@10 of DP-FedNeuMF and FedNeuMF models are different, but they will continue to increase, and then gradually stabilize, while the loss rate loss is different. The same, but both keep decreasing and then gradually stabilize.

To sum up, after adding differential privacy, we add noise to the data on the basis of ensuring the recommendation effect and training efficiency, which preserves the user's privacy and causes PPA more difficult. It can be considered that the algorithm after adding differential privacy has achieved our expected goal.

5 Conclusion

In this paper, a neural matrix factorization recommendation model based on federated learning and a gradient transmission model based on differential privacy are designed. The training efficiency and recommendation effect of three examples of federated neural collaborative filtering are compared and analyzed through experiments. The experimental results show that the federated neural matrix factorization algorithm has better training efficiency and recommendation effect than the other two algorithms. The training efficiency and recommendation effect of federated neural matrix decomposition before and after adding differential privacy are compared and analyzed through experiments. The experimental results show that the training efficiency and recommendation effect of the algorithm after adding differential privacy are not greatly affected. The improved algorithm is feasible.

Of course, there is still room for improvement in the current work. For example, the recommendation algorithm in this paper uses neural matrix decomposition, one of the examples of neural collaborative filtering. This paper only

compares it with the federation of the other two examples, and does not compared with some more advanced recommendation algorithms that currently exist; secondly, although this paper can reduce the probability of being successfully attacked by preference analysis by adding differential privacy, it also increases the computing cost of the client and the communication between the client and the central server. These are not considered at present; in addition, there are many types of attack threats faced by federated learning, and this article only considers one of them, the PPA.

Therefore, in future work, the author will conduct further research on how to choose a recommendation algorithm to improve the recommendation effect, how to improve the algorithm to resist PPA while reducing computing and communication costs, and how to resist more types of attacks.

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